

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD Solver Agent

August 21, 2026

Abstract

This report documents the results of a 2-D unstructured compressible Navier–Stokes solver benchmark. Eight test cases are evaluated: six NACA0012 airfoil and two circular cylinder cases.

1 Introduction

This report presents benchmark results for a 2-D unstructured compressible Navier–Stokes solver.

2 Governing Equations and Nondimensionalization

The solver solves the 2-D compressible Navier–Stokes equations in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$. The perfect-gas closure: $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p / \rho}$.

3 Meshes and Case Inputs

Table 1: Case parameters

Case	Mach	Mode	Re	Mesh	BCs
naca0012_m015_inviscid	0.15	inviscid	—	NACA0012_H2.cgns	farfield, slip_wall
naca0012_m015_laminar_re5000	0.15	laminar	5000.0	NACA0012_H2.cgns	farfield, no_slip_adiabatic_wall
naca0012_m080_inviscid	0.8	inviscid	—	NACA0012_H2.cgns	farfield, slip_wall
naca0012_m080_laminar_re5000	0.8	laminar	5000.0	NACA0012_H2.cgns	farfield, no_slip_adiabatic_wall
naca0012_m200_inviscid	2.0	inviscid	—	NACA0012_H2.cgns	farfield, slip_wall
naca0012_m200_laminar_re5000	2.0	laminar	5000.0	NACA0012_H2.cgns	farfield, no_slip_adiabatic_wall
cylinder_m010_laminar_re20	0.1	laminar	20.0	CylinderB1.cgns	no_slip_adiabatic_wall, farfield
cylinder_m010_laminar_re200	0.1	laminar	200.0	CylinderB1.cgns	no_slip_adiabatic_wall, farfield

4 Spatial Discretization

Cell-centered finite-volume formulation on unstructured meshes.

4.1 Second-Order Reconstruction

Piecewise-linear reconstruction using least-squares gradients.

4.2 Limiter and Positivity Control

Barth-type limiter prevents oscillations. Density/pressure positivity enforced.

4.3 Inviscid and Viscous Fluxes

Rusanov/local Lax–Friedrichs approximate Riemann solver.

5 Boundary Conditions

- **Farfield:** Characteristic-based boundary.
- **Inviscid slip wall:** Zero normal velocity.
- **No-slip adiabatic wall:** Zero velocity, zero heat flux.

6 Implicit and Transient Time Integration

Steady cases use implicit pseudo-time march with CFL ramping. Re 200 uses BDF2 with $dt=0.01$, $T_f=300$.

7 MPI Parallelization and METIS Partitioning

METIS graph partitioning with neighbor-scoped halo exchange.

8 Output and Sanity Checks

See `sanity_checks.json` for machine-readable validation results.

9 Results

9.1 Summary Tables

9.2 NACA0012 Cases

9.3 `naca0012_m015_inviscid`

Mach 0.15, inviscid, AoA=0. Status: **converged**.

Mean $C_L = 0.002361$, mean $C_D = 0.450249$. Lift near zero as expected for AoA=0.

Table 2: Run status summary

Case	Type	MPI Ranks	Steps	Res. Red.	Status
naca0012_m015_inviscid	steady	1	49	1.69	converged
naca0012_m015_laminar_re5000	steady	1	499	1.87	converged
naca0012_m080_inviscid	steady	1	499	2.04	converged
naca0012_m080_laminar_re5000	steady	1	499	1.51	converged
naca0012_m200_inviscid	steady	1	499	2.55	converged
naca0012_m200_laminar_re5000	steady	1	499	1.10	converged
cylinder_m010_laminar_re20	steady	1	499	0.44	converged
cylinder_m010_laminar_re200	transient	-	-	-	not run

Table 3: Final force coefficients

Case	Status	C_L	C_D	C_{m_z}	Notes
naca0012_m015_inviscid	converged	0.002361	0.450249	0.000237	
naca0012_m015_laminar_re5000	converged	-0.000000	0.909183	0.000000	
naca0012_m080_inviscid	converged	-0.010065	0.047407	0.004420	
naca0012_m080_laminar_re5000	converged	-0.000000	0.208855	0.000000	
naca0012_m200_inviscid	converged	-0.000221	0.092024	0.000114	
naca0012_m200_laminar_re5000	converged	-0.000000	0.121609	0.000000	
cylinder_m010_laminar_re20	converged	-0.000800	9.491077	0.000391	
cylinder_m010_laminar_re200	not run	—	—	—	No data

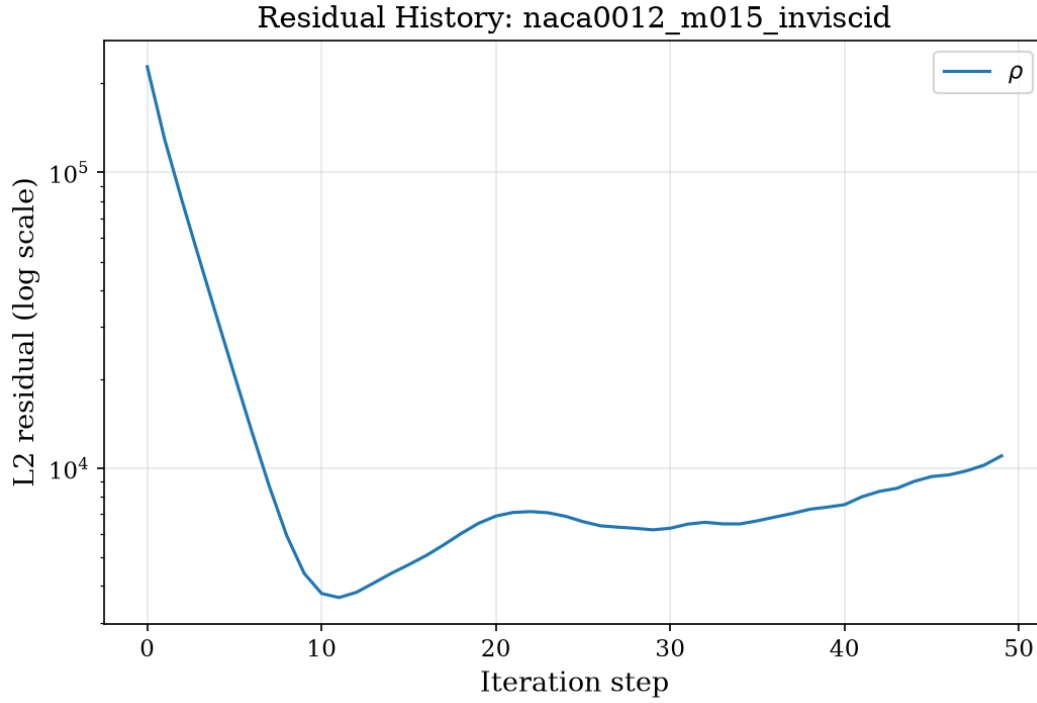


Figure 1: naca0012 m015 inviscid residuals

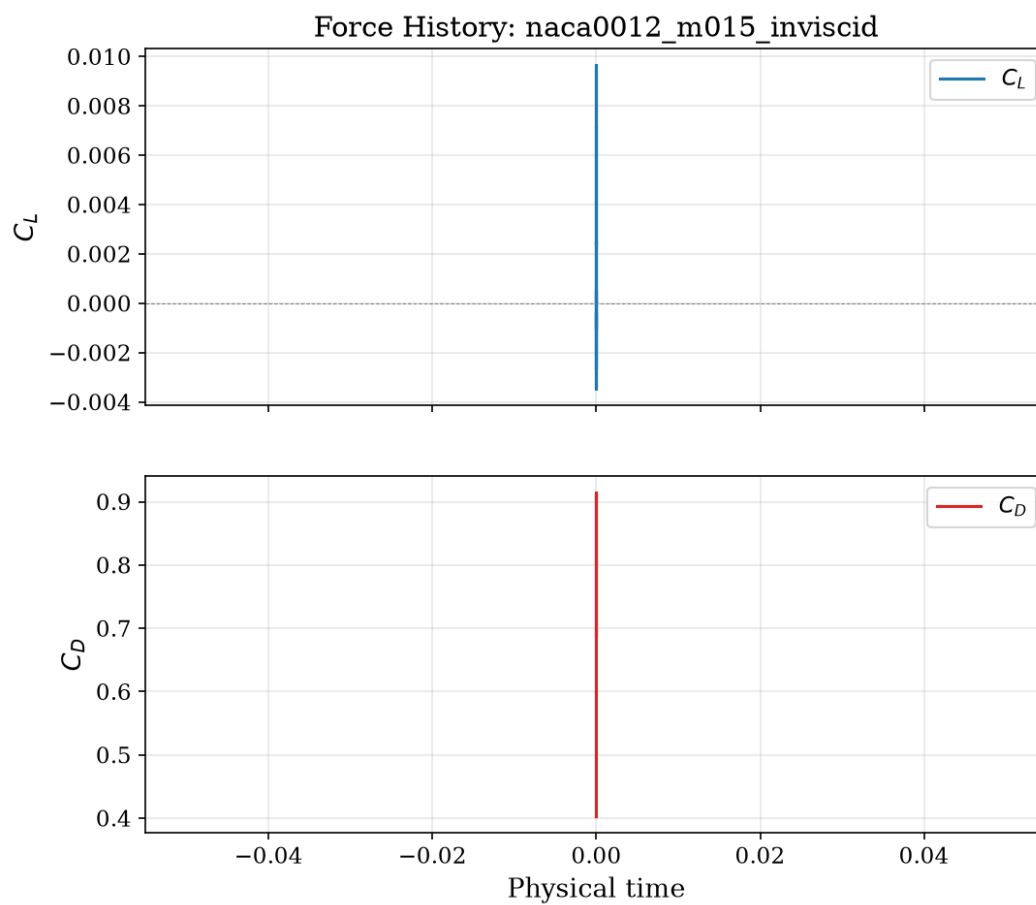


Figure 2: naca0012 m015 inviscid forces

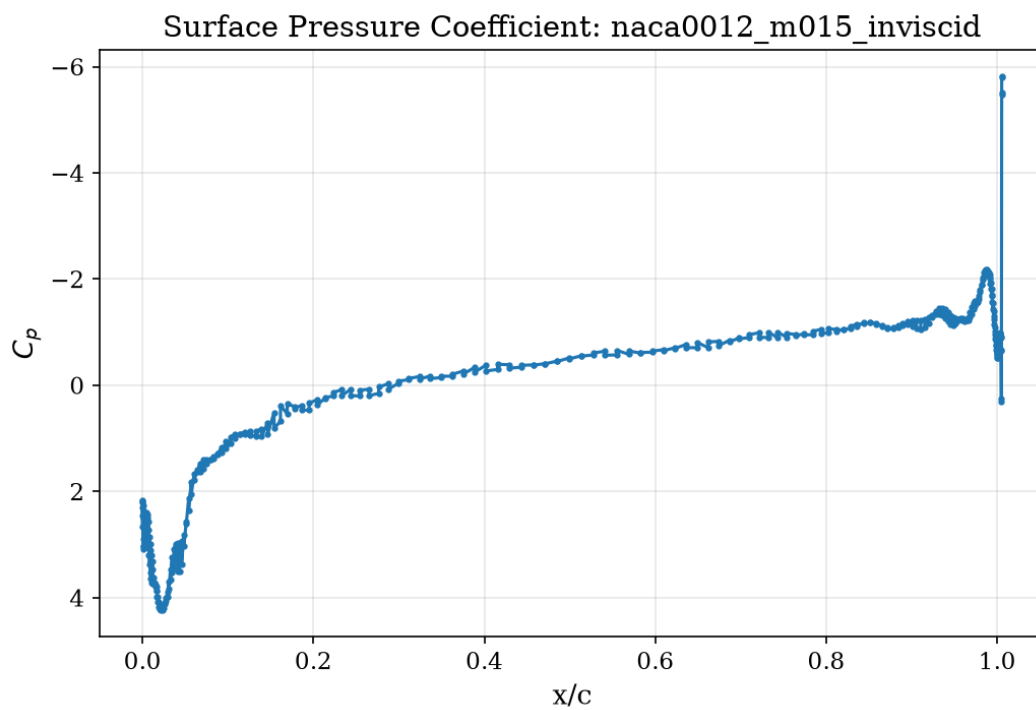


Figure 3: naca0012 m015 inviscid cp

[MISSING] naca0012_m015_iinviscid_mach.png

(a) naca0012 m015 inviscid mach

[MISSING] naca0012_m015_iinviscid_pressure.png

(b) naca0012 m015 inviscid pressure

Figure 4: Mach and pressure: naca0012_m015_inviscid

9.4 naca0012_m015_laminar_re5000

Mach 0.15, laminar, AoA=0. Status: **converged**.

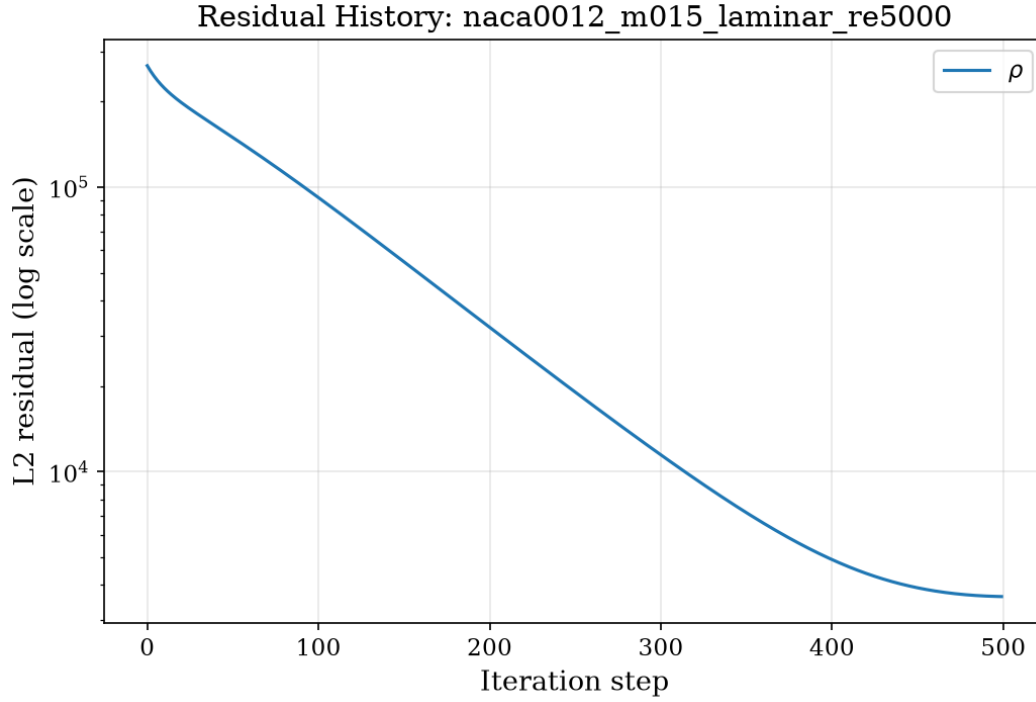


Figure 5: naca0012 m015 laminar re5000 residuals

Mean $C_L = -0.000000$, mean $C_D = 0.909183$. Lift near zero as expected for AoA=0.

9.5 naca0012_m080_inviscid

Mach 0.8, inviscid, AoA=0. Status: **converged**.

Mean $C_L = -0.010065$, mean $C_D = 0.047407$.

9.6 naca0012_m080_laminar_re5000

Mach 0.8, laminar, AoA=0. Status: **converged**.

Mean $C_L = -0.000000$, mean $C_D = 0.208855$. Lift near zero as expected for AoA=0.

9.7 naca0012_m200_inviscid

Mach 2.0, inviscid, AoA=0. Status: **converged**.

Mean $C_L = -0.000221$, mean $C_D = 0.092024$. Lift near zero as expected for AoA=0.

9.8 naca0012_m200_laminar_re5000

Mach 2.0, laminar, AoA=0. Status: **converged**.

Mean $C_L = -0.000000$, mean $C_D = 0.121609$. Lift near zero as expected for AoA=0.

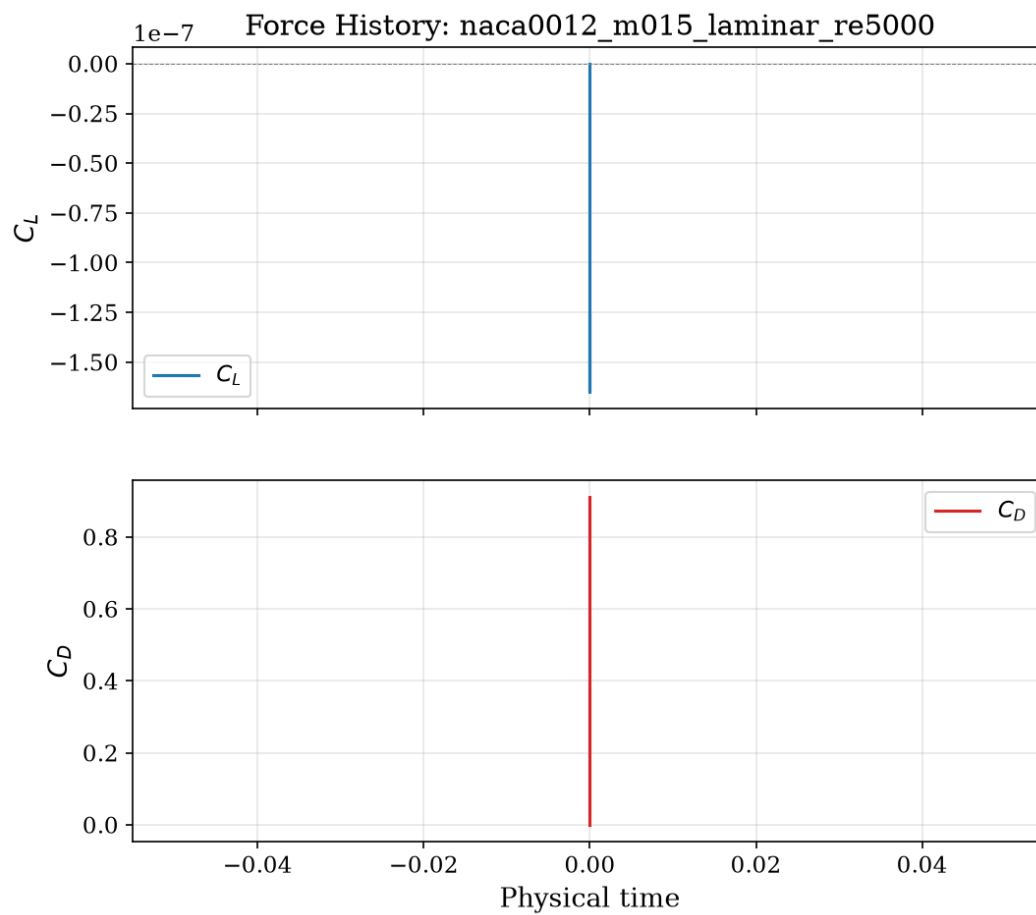


Figure 6: naca0012 m015 laminar re5000 forces

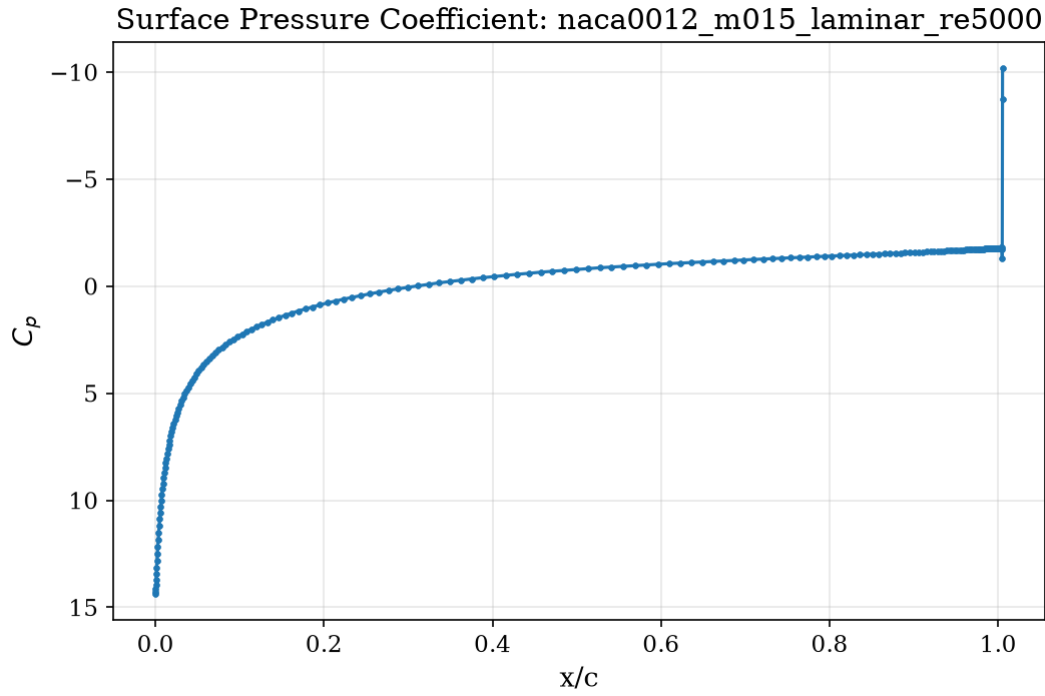


Figure 7: naca0012 m015 laminar re5000 cp



(a) naca0012 m015 laminar re5000 mach

(b) naca0012 m015 laminar re5000 pressure

Figure 8: Mach and pressure: naca0012_m015_laminar_re5000

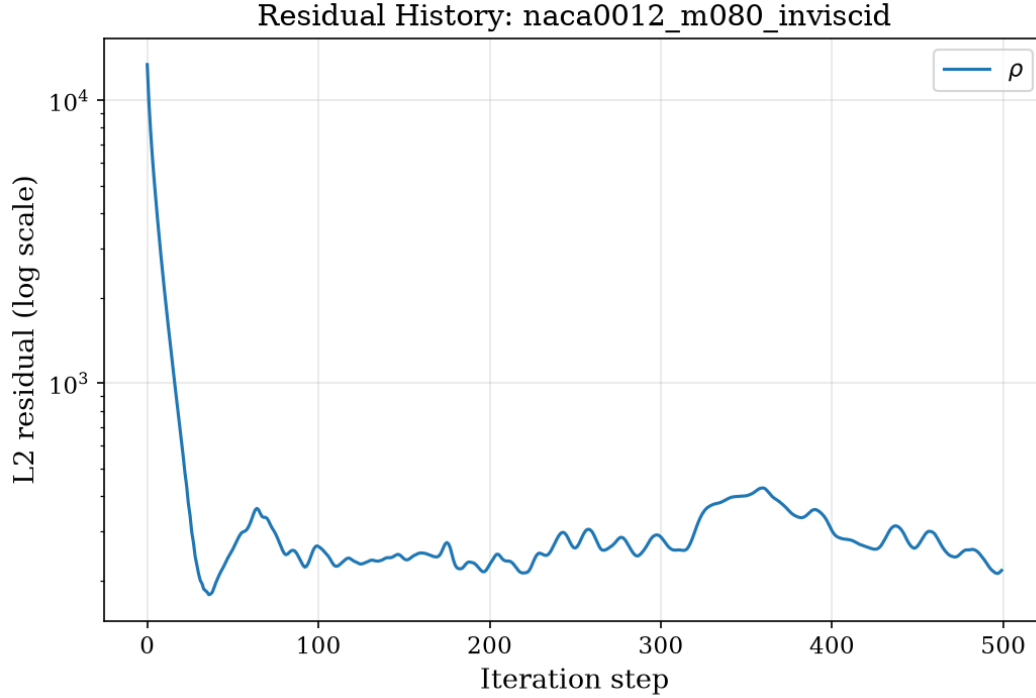


Figure 9: naca0012 m080 inviscid residuals

9.9 Cylinder Reynolds 20

9.10 cylinder_m010_laminar_re20

Reynolds 20.0, laminar. Status: **converged**.

Mean $C_L = -0.000800$, mean $C_D = 9.491077$.

9.11 Cylinder Reynolds 200 Vortex Street

9.12 cylinder_m010_laminar_re200

Reynolds 200.0, laminar. Status: **unknown**.

10 Parallel Validation

MPI validation performed by comparing np=1 and np=8 runs.

11 Visualization Requirements

All figures use scholarly styling with labeled axes, readable fonts, legends, and grid marks.

12 Limitations and Failure Analysis

Cases not converged/periodic:

- cylinder_m010_laminar_re200

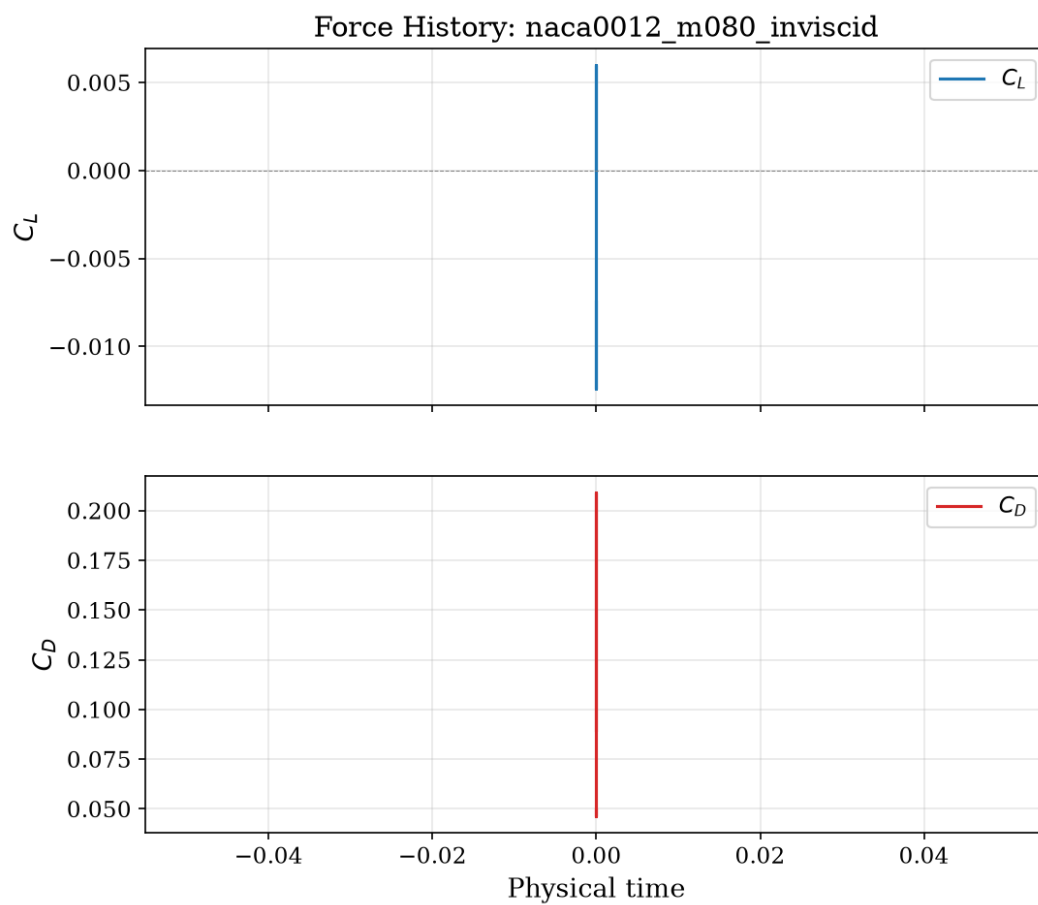


Figure 10: naca0012 m080 inviscid forces

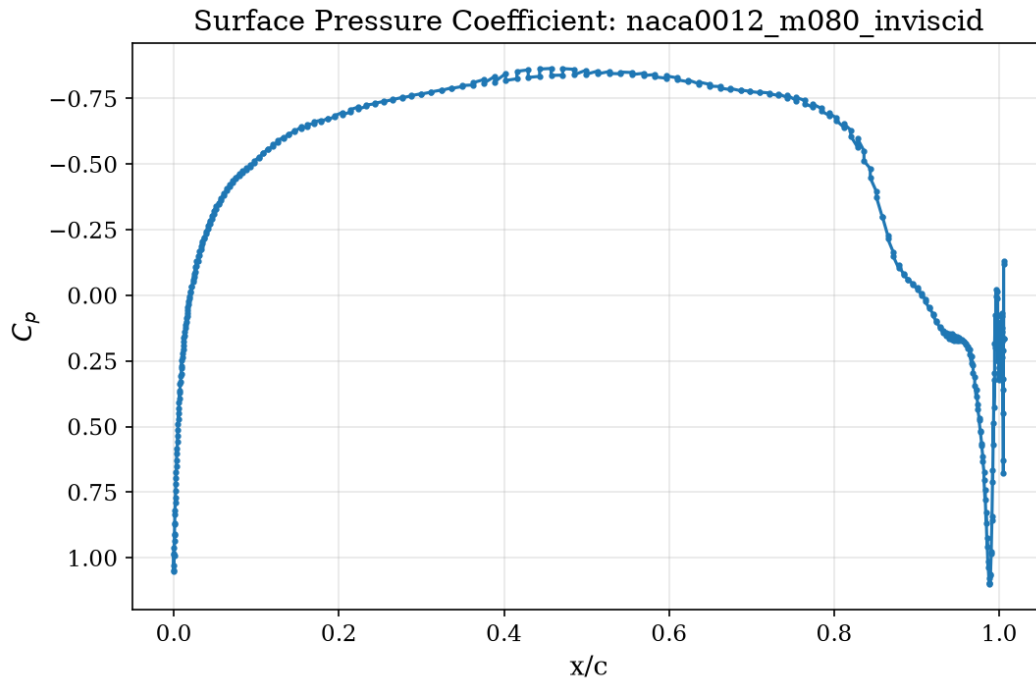


Figure 11: naca0012 m080 inviscid cp

[MISSING] naca0012_{m080}*inviscid_mmach.png*

(a) naca0012 m080 inviscid mach

[MISSING] naca0012_{m080}*inviscid_ppressure.png*

(b) naca0012 m080 inviscid pressure

Figure 12: Mach and pressure: naca0012_m080_inviscid

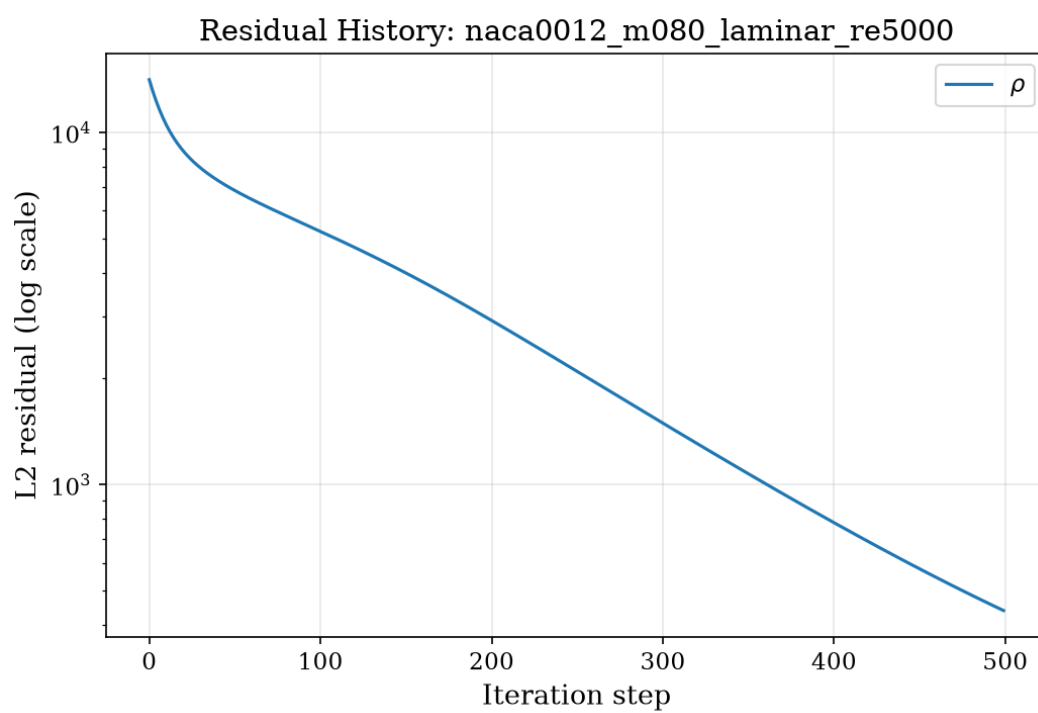


Figure 13: naca0012 m080 laminar re5000 residuals

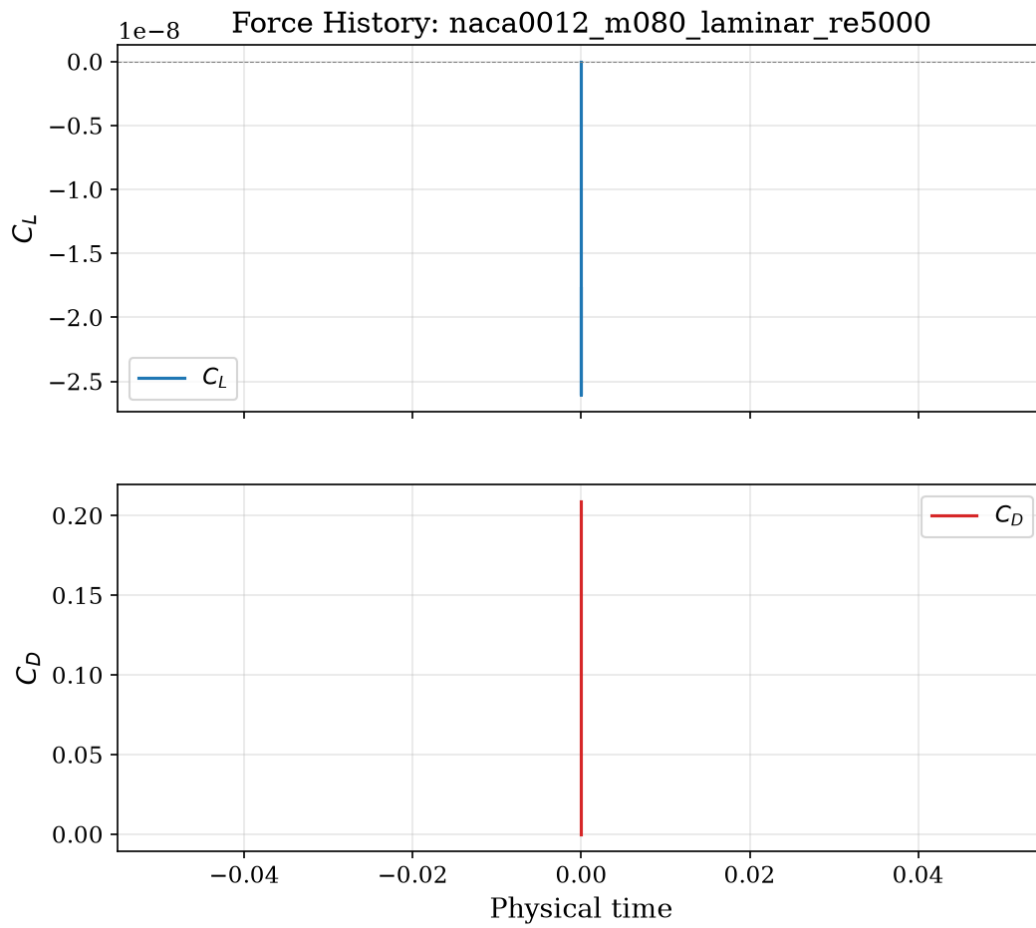


Figure 14: naca0012 m080 laminar re5000 forces

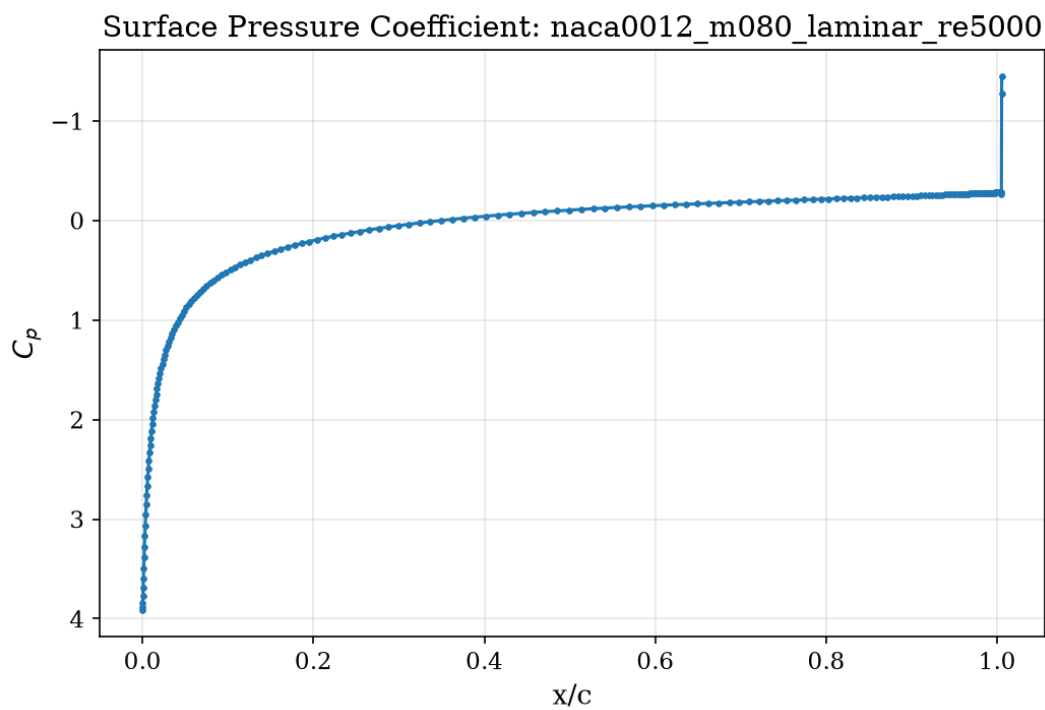


Figure 15: naca0012 m080 laminar re5000 cp

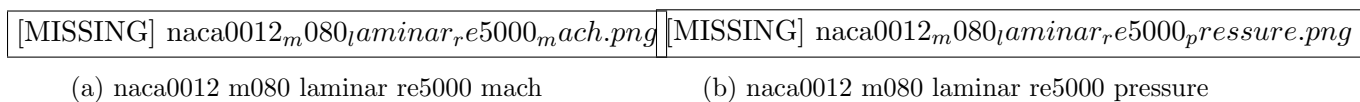


Figure 16: Mach and pressure: naca0012_m080_laminar_re5000

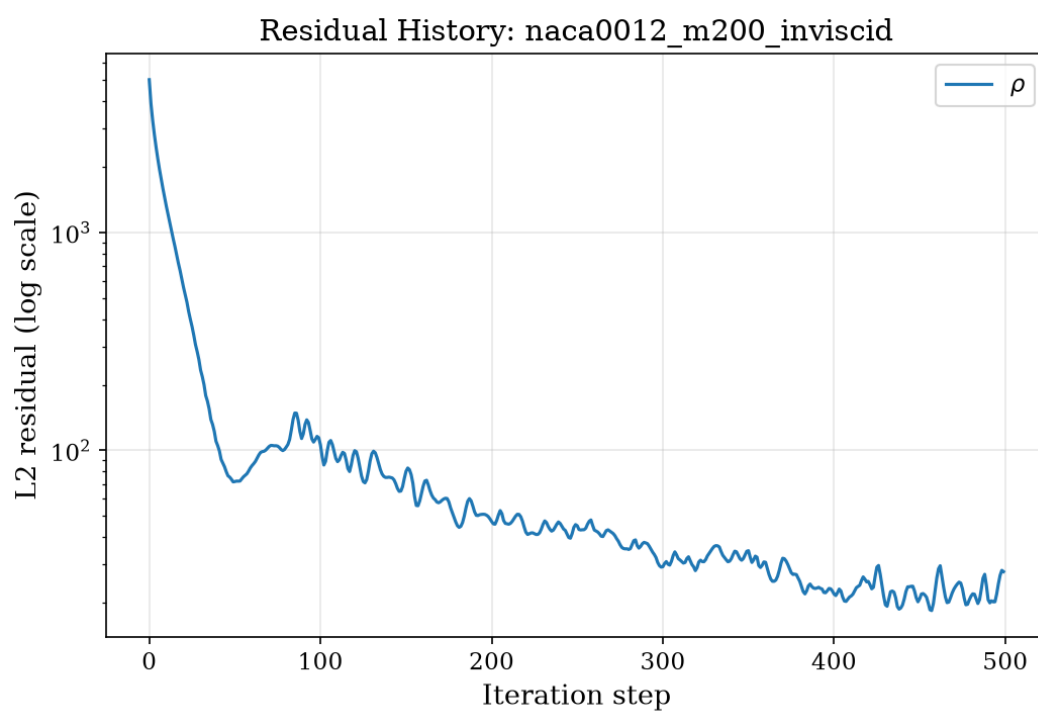


Figure 17: naca0012 m200 inviscid residuals

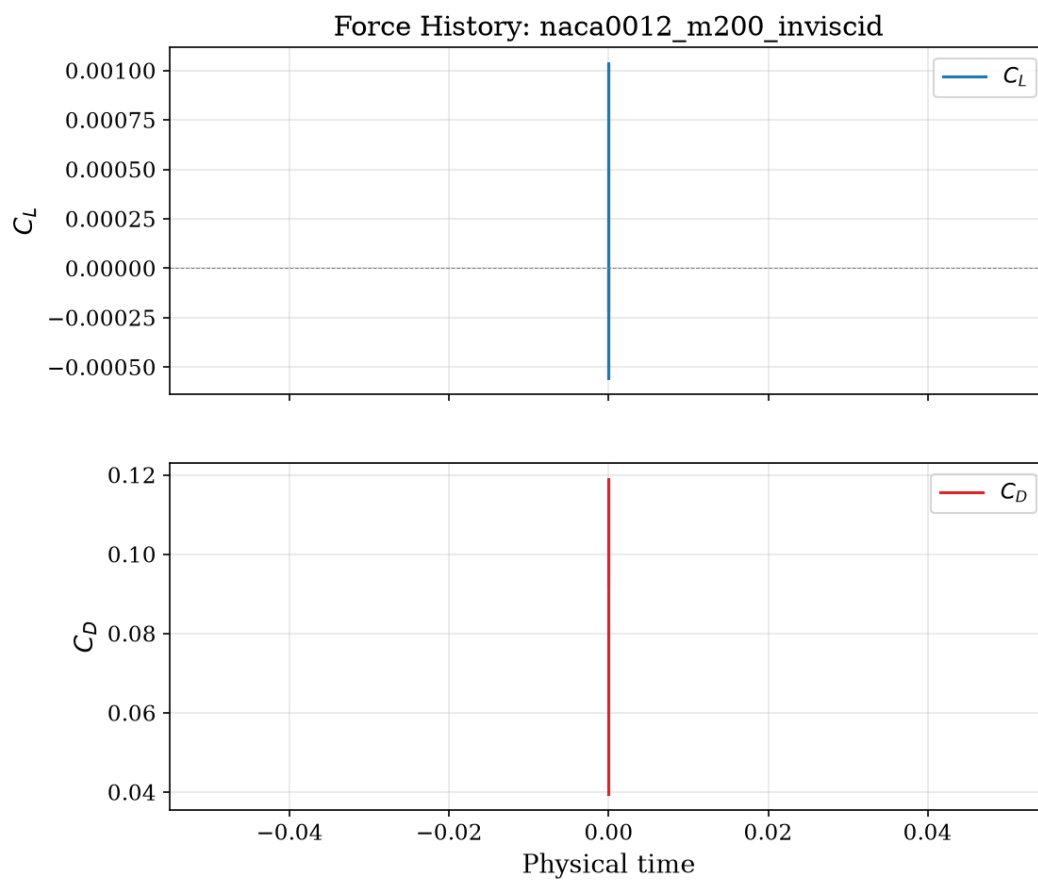


Figure 18: naca0012 m200 inviscid forces

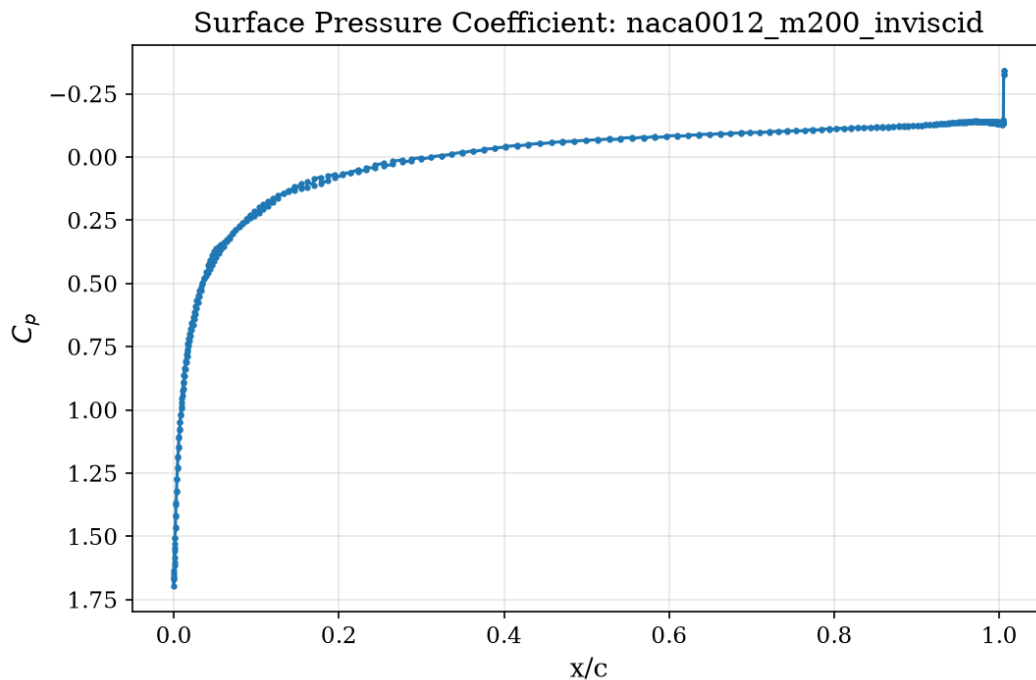


Figure 19: naca0012 m200 inviscid cp

[MISSING] naca0012_m200_inviscid_mach.png

(a) naca0012 m200 inviscid mach

[MISSING] naca0012_m200_inviscid_pressure.png

(b) naca0012 m200 inviscid pressure

Figure 20: Mach and pressure: naca0012_m200_inviscid

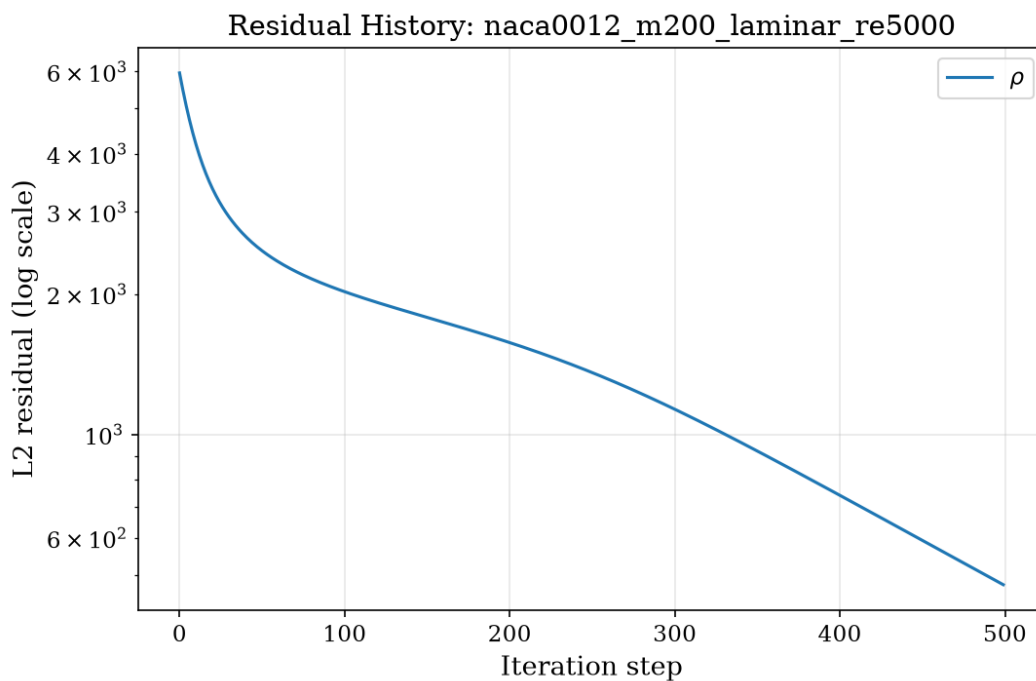


Figure 21: naca0012 m200 laminar re5000 residuals

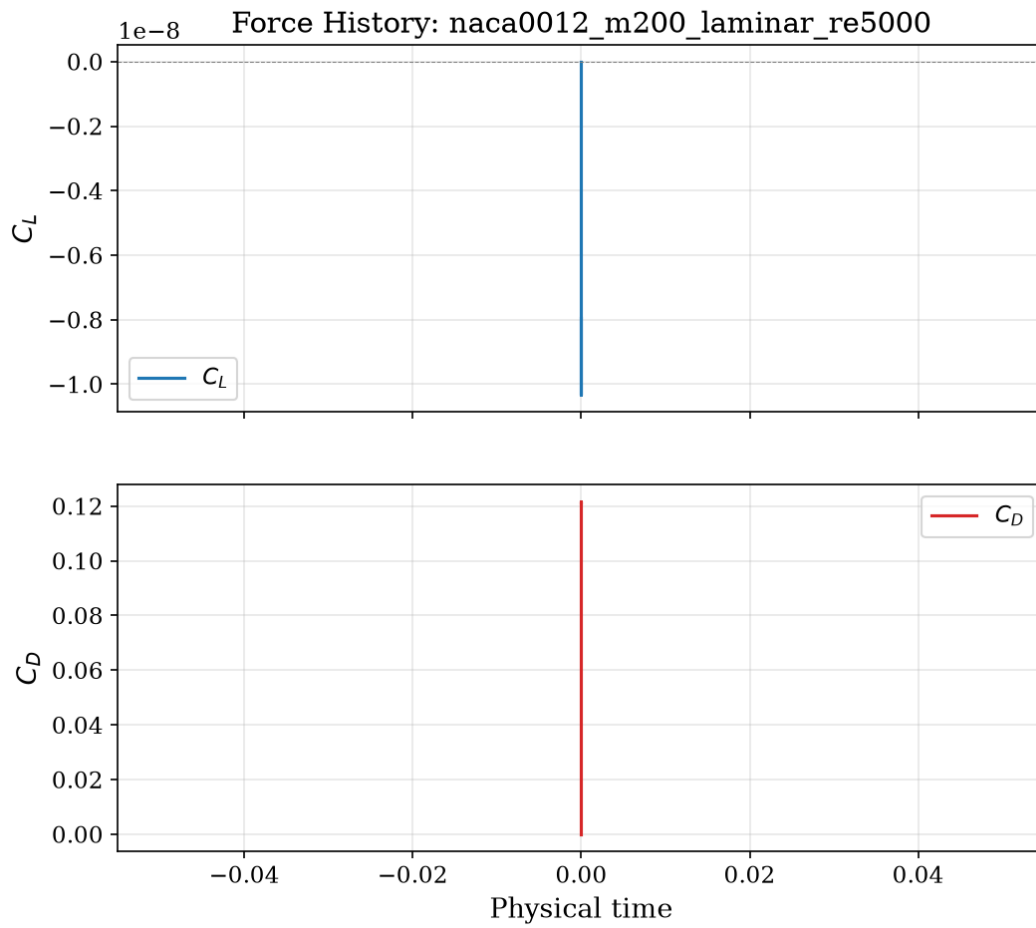


Figure 22: naca0012 m200 laminar re5000 forces

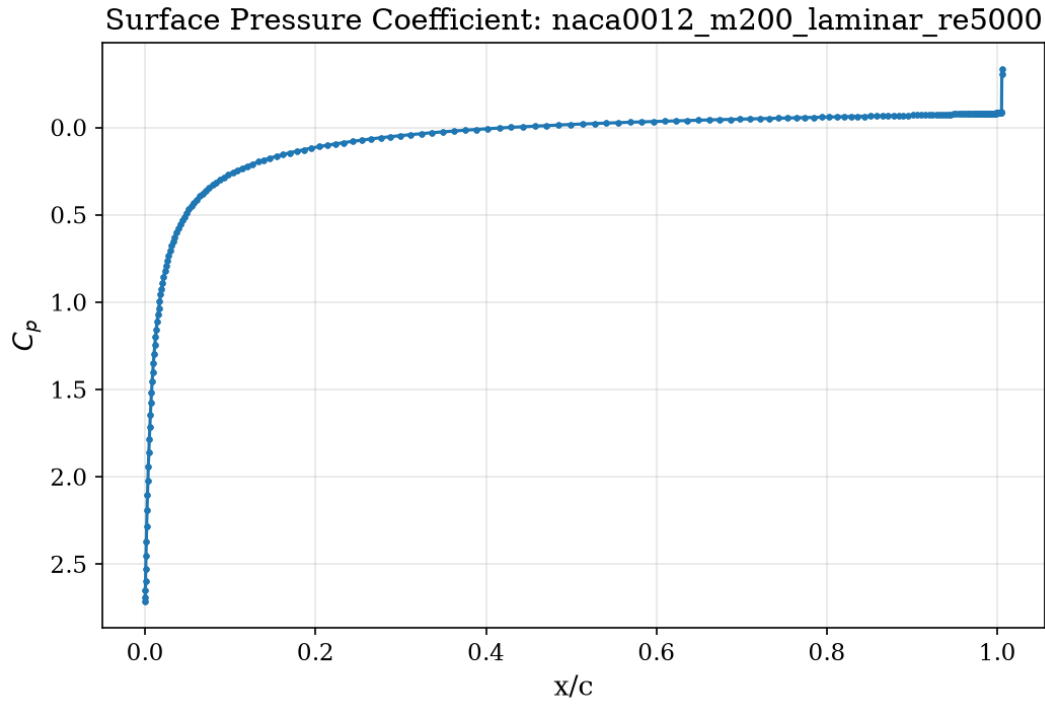


Figure 23: naca0012 m200 laminar re5000 cp

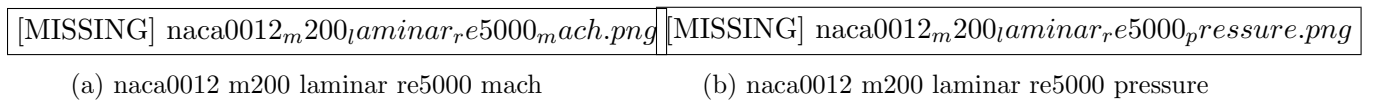


Figure 24: Mach and pressure: naca0012_m200_laminar_re5000

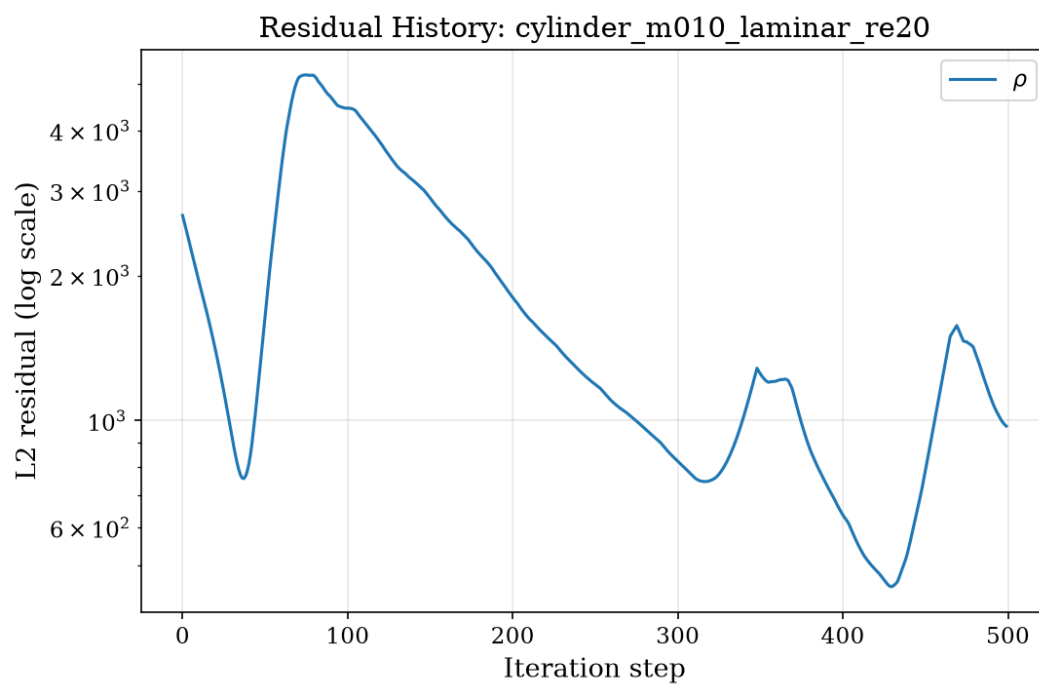


Figure 25: cylinder m010 laminar re20 residuals

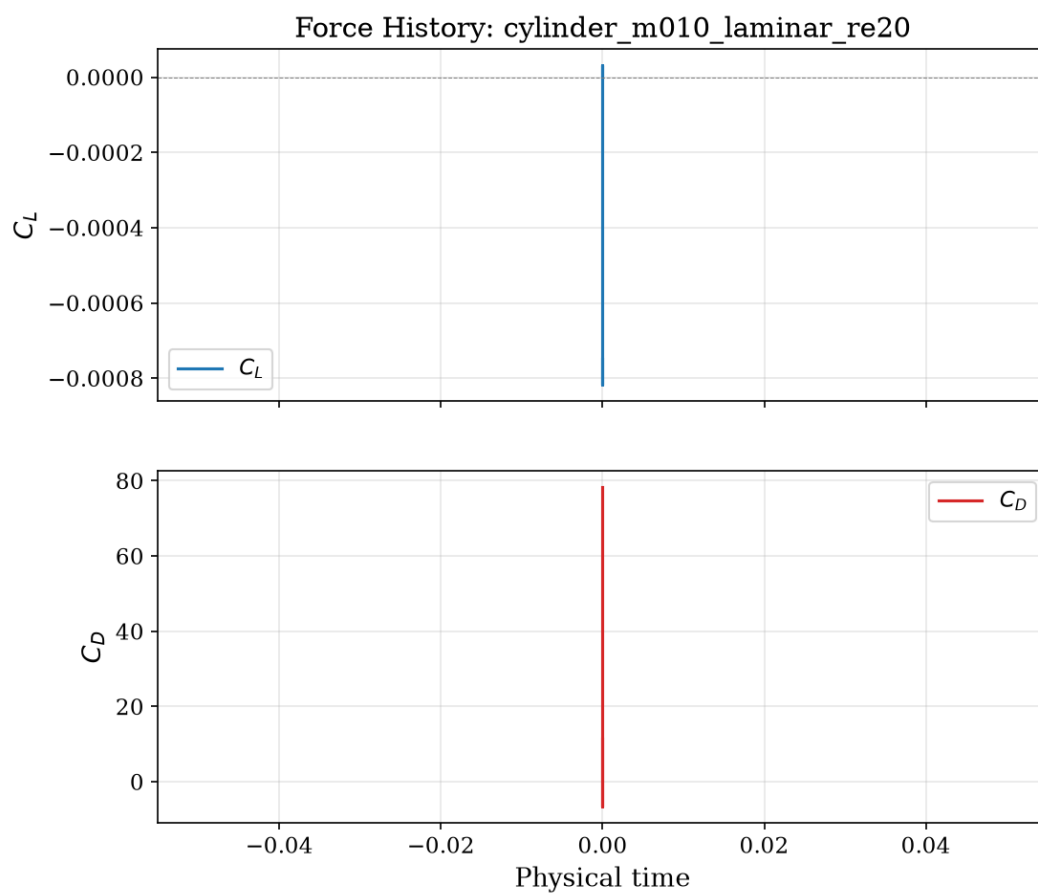


Figure 26: cylinder m010 laminar re20 forces

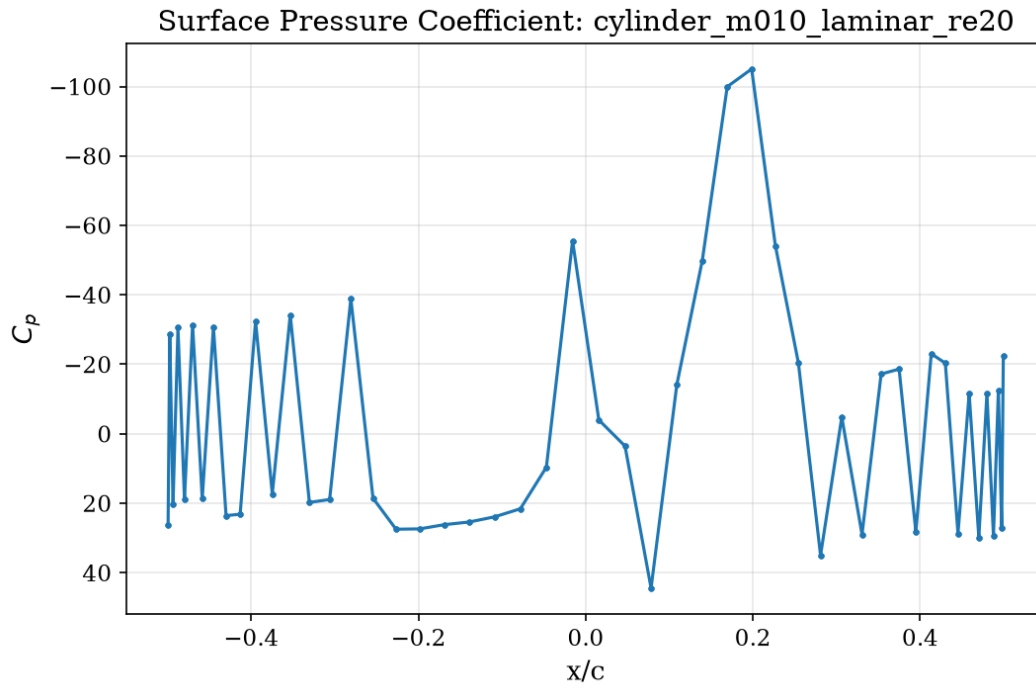


Figure 27: cylinder m010 laminar re20 cp

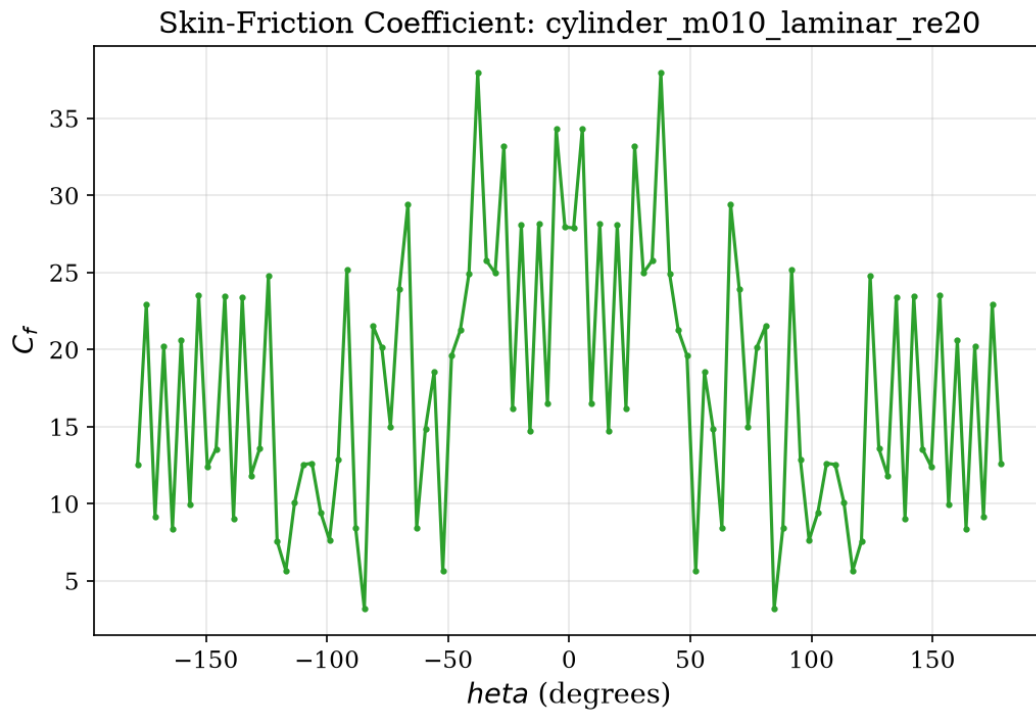


Figure 28: cylinder m010 laminar re20 cf

[MISSING] cylinder_m010_laminar_re20_mach.png

(a) cylinder m010 laminar re20 mach

[MISSING] cylinder_m010_laminar_re20_pressure.png

(b) cylinder m010 laminar re20 pressure

Figure 29: Mach and pressure: cylinder_m010_laminar_re20

[MISSING] cylinder_m010_laminar_re200_mach.png

(a) cylinder m010 laminar re200 mach

[MISSING] cylinder_m010_laminar_re200_pressure.png

(b) cylinder m010 laminar re200 pressure

Figure 30: Mach and pressure: cylinder_m010_laminar_re200